

Educational Scenarios

**Select
Scenario:**

Scenario 1: Perfect Secrecy (Proper Usage)



Perfect One-Time Pad (Secure)

True one-time pad with random key same length as message

Try the encryption yourself with the scenario values:



Plaintext:

11010011



Key:

01101100



Expected

10111111

Result:



Security

Status:



Secure (Perfect Secrecy)

Your

Result:

10111111



Encrypt



Test Your Understanding

Predict

Ciphertext:

01010001100010

Security

Assessment:

Insecure (Vulnerable)



 Check My Answer

 Learning Feedback

✗ **Incorrect ciphertext.** Expected: 10111111, You entered: 01010001100010

✗ **Incorrect security assessment.** This scheme is **secure**

 Explanation:

✓ Secure: Random key, same length as message, used only once. This achieves perfect secrecy.

 Learning Point:

One-time pad with proper key achieves perfect secrecy

 Correct Answers:

Ciphertext: 10111111

Security: secure

 Vernam Cipher Simulator

 **Plaintext:**

10010011000110

 **Key:**

11000010100100

 Generate Random Key



Ciphertext:

01010001100010

 Encrypt ↓

 Decrypt ↑